

Last time f unif. cts $\Leftrightarrow \forall (a_n)$ Cauchy, $(f(a_n))$ Cauchy

$\Rightarrow$ OK

$\Leftarrow$ Doesn't work

Ex. $f(x) = x^2$ then we know from a previous class that f not unif. cts.
But we'll show that it preserves Cauchy Sequences.

Fact. Cauchy seqs are bounded

Quick PF Since Cauchy seqs converge $(a_n) \rightarrow M$

then let $\varepsilon = 1$

$\exists N$ s.t. $\forall n \geq N, M-1 \leq a_n \leq M+1$

(a_n) is bounded by $\max\{a_1, a_N, M+1\}$.

Let (a_n) be Cauchy

WTS: $(f(a_n)) = (a_n^2)$ is Cauchy

i.e. $\forall \varepsilon > 0, \exists N$ s.t. $\forall n, m \geq N$

$$|a_n^2 - a_m^2| < \varepsilon.$$

$$|a_n^2 - a_m^2| = |a_n + a_m| + |a_n - a_m|$$

$$\overset{a_n \text{ bounded}}{\leq} (|a_n + a_m|) |a_n - a_m|$$

$$\text{by } B \rightarrow \leq 2B |a_n - a_m| \overset{\text{want}}{<} \varepsilon$$

We know (a_n) is Cauchy, so for $\varepsilon/2B \exists N$ s.t. $\forall n, m \geq N$

$$|a_n - a_m| < \varepsilon/2B$$

For this N , $\forall n, m \geq N$

$$\text{We have } |a_n^2 - a_m^2| < 2\delta \cdot \frac{\varepsilon}{2\delta} = \varepsilon$$



Actual Sequential Criterion for Uni Continuity

Thm f unif. cts.

$$\Leftrightarrow \forall (x_n), (y_n) \text{ sequences s.t. } \lim_{n \rightarrow \infty} (x_n - y_n) = 0$$

$$\text{We have } \lim_{n \rightarrow \infty} (f(x_n) - f(y_n)) = 0$$

PF $(\Rightarrow)$ Assume f unif. cts.

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, y \text{ w/ } |x - y| < \delta, |f(x) - f(y)| < \varepsilon.$$

$$\text{WTS. If } \lim_{n \rightarrow \infty} (x_n - y_n) = 0$$

$$\text{Want } \left[\begin{array}{l} \text{then } \lim_{n \rightarrow \infty} (f(x_n) - f(y_n)) = 0 \\ \forall \varepsilon > 0, \exists N \text{ s.t. } \forall n \geq N, |f(x_n) - f(y_n)| < \varepsilon \end{array} \right.$$

$$\text{Let } N \text{ be s.t. } \forall n \geq N, |x_n - y_n| < \delta$$

$$\text{Exists b/c. } \lim_{n \rightarrow \infty} (x_n - y_n) = 0$$

$$\text{Then by uni. continuity, } \forall n \geq N, |f(x_n) - f(y_n)| < \varepsilon$$



$(\Leftarrow)$. Suppose for a contradiction that f not unif. cts.

$$\exists \varepsilon > 0 \quad \forall \delta > 0 \quad \exists x, y \text{ w/ } |x - y| < \delta \text{ s.t. } |f(x) - f(y)| \geq \varepsilon$$

$$\text{WTS. } \exists \text{ seq. } (x_n), (y_n) \text{ w/ } \lim_{n \rightarrow \infty} (x_n - y_n) = 0$$

$$\text{but } \lim_{n \rightarrow \infty} (f(x_n) - f(y_n)) \neq 0.$$

Take the ε from the fact that f unif. lts.

Construct $(x_n), (y_n)$ by the following process

- $\delta = 1 : \exists x_1, y_1$ w/ $|x_1 - y_1| < 1$, but $|f(x_1) - f(y_1)| \geq \varepsilon$
- $\delta = 1/2 : \exists x_2, y_2$ w/ $|x_2 - y_2| < 1/2$, but $|f(x_2) - f(y_2)| \geq \varepsilon$

Then $|x_n - y_n| < \frac{1}{n} \Rightarrow \lim_{n \rightarrow \infty} (x_n - y_n) = 0$

but $\lim_{n \rightarrow \infty} |f(x_n) - f(y_n)| \neq 0$ Contradiction.



Rmk We don't need $(x_n), (y_n)$ to be convergent.

But if f has a bounded domain, we can even require $(x_n), (y_n)$ convergent.

- Use Bolzano-Weierstrass on the bounded seq (x_n) to get $(x_{n_i})_i$, convergent subseq.

Now apply BW on $(y_{n_i})_i$ & find convergent $(y_{n_{i_j}})_j$
 $\Rightarrow (x_{n_{i_j}})_j, (y_{n_{i_j}})_j$ will work

IX. Derivatives

Def A function f is differentiable at a point a if $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists.

This limit is the derivative of f at a . $f'(a)$

We say that f is differentiable if it is differentiable at every pt. in its domain.

$$f: D \rightarrow \mathbb{R}$$

$$f': D \rightarrow \mathbb{R}$$

$$a \mapsto f'(a)$$

Ex ① If $s(t)$ = position of some particle moving on the x-axis at time t .

$$\text{Then at time } a, \quad s'(a) = \lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h}$$

= instantaneous Velocity

② If $f(x) = c$, constant function

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c - c}{h} = 0 \quad \forall a \Rightarrow f'(x) = 0 \end{aligned}$$

③ $f(x) = cx + d$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = c \quad \forall a \Rightarrow f'(x) = c$$

$$\textcircled{4} \quad f'(x^n) = nx^{n-1}$$

⑤ If $f(x) = 3x^2 + 5$

What is the equation of the tangent line through $(a, 3a^2 + 5)$?

$$\frac{f(a+h)-f(a)}{h} = \frac{\Delta y}{\Delta x}$$

= slope at a

Looking for equation of the line w/ slope $f'(a) = 6a$.

Passes through $(a, 3a^2+5)$

$$y = (6a)x + b, \text{ solve for } b$$

$$3a^2+5 = 6a^2+b \Rightarrow b = -3a^2+5$$

$$y = 6ax + (-3x^2+5)$$

$$\textcircled{6} \text{ If } f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Then $f(x)$ is not differentiable at $x=0$

* Suppose the limit $\lim_{x \rightarrow a} \frac{f(a+h)-f(a)}{h}$ exists = l .

For $\varepsilon > 0$, should find $\delta > 0$ s.t.

$$\text{whenever } |h| < \delta, \left| \frac{f(a+h)-f(a)}{h} \right| < \varepsilon$$

$$\text{Use } |f(a+h)-f(a)| < 2, \text{ so } \left| \frac{f(a+h)-f(a)}{h} - l \right| \leq \left| \frac{2}{h} \right| + l$$